AS · Cambridge (CIE) · Physics

22 mins 22 questions

Multiple Choice Questions

Errors & Uncertainties

Errors & Uncertainties / Calculating Uncertainties

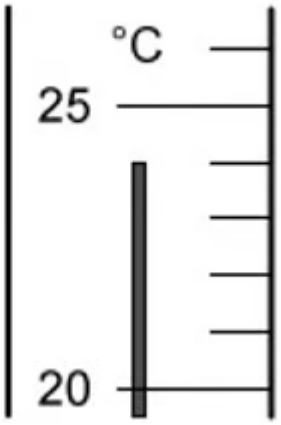
Easy (6 questions)	/6
Medium (8 questions)	/8
Hard (8 questions)	/8
Total Marks	**/22**

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Easy Questions

1 The diagram shows part of a thermometer.



What is the correct reading on the thermometer and the uncertainty in this reading?

	Reading/°C	Uncertainty in reading/°C
A	24	±1
B	24	±0.1
C	24.0	±0.2
D	24.0	±0.5

(1 mark)

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2 In an experiment to determine the acceleration of free fall using a falling body, what would lead to a value that is too large?

A. air resistance

B. dimensions of the body are too large

C. measured distance longer than true distance

D. measured time longer than true-time

(1 mark)

3 A metre rule is used to measure the length of a piece of wire. It is found to be 70 cm long to the nearest millimetre.

How should this result be recorded in a table of results?

A. 0.7 m

B. 0.70 m

C. 0.700 m

D. 0.7000 m

(1 mark)

4 An experiment is done to measure the acceleration of free fall of a body from rest.

Which measurements are needed?

A. The height of fall and the time of fall

B. The height of fall and the weight of the body

C. The mass of the body and the height of fall

D. The mass of the body and the time of fall

(1 mark)

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5 A micrometer is used to measure the diameters of two cylinders

diameter of first cylinder = 12.78 ± 0.02 mm

diameter of second cylinder = 16.24 ± 0.03 mm

The difference in the diameters is calculated.

What is the uncertainty in this difference?

A. ± 0.01 mm

B. ± 0.02 mm

C. ± 0.03 mm

D. ± 0.05 mm

(1 mark)

6 A student carries out a series of determinations of the acceleration of free fall *g*. The table shows the results.

g / m s^{-2}
4.91
4.89
4.88
4.90
4.93
4.92

What can be said about this experiment?

A. it is accurate and precise

B. it is accurate but not precise

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C. it is not accurate and not precise

D. it is not accurate but is precise

(1 mark)

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Medium Questions

1 The measurement of a physical quantity may be subject to random errors and systematic errors.

Which statement is correct?

A. random errors can be reduced by taking the average of several measurements

B. random errors are always caused by the person taking the measurement

C. a systematic error cannot be reduced by adjusting the apparatus

D. a systematic error results in a different reading each time the measurement is taken

(1 mark)

2 In an experiment, a radio-controlled car takes 2.50 ± 0.05 s to travel 40.0 ± 0.1 m.

What is the car's average speed and the uncertainty in this value?

A. 16 ± 1 m s^{-1}

B. 16.0 ± 0.2 m s^{-1}

C. 16.0 ± 0.4 m s^{-1}

D. 16.0 ± 0.36 m s^{-1}

(1 mark)

3 Four students each made a series of measurements of the acceleration of free fall *g*. The table shows the results obtained.

Which set of results could be described as precise but not accurate?

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	g / ms^{-2}			
A	9.81	9.79	9.84	9.83
B	9.81	10.12	9.89	9.94
C	9.45	9.21	8.99	8.76
D	8.45	8.46	8.50	8.41

(1 mark)

4 Which experimental technique reduces the systematic error of the quantity being investigated?

A. adjusting an ammeter to remove its zero error before measuring a current

B. measuring several internodal distances on a standing wave to find the mean internodal distance

C. measuring the diameter of a wire repeatedly and calculating the average

D. timing a large number of oscillations to find a period

(1 mark)

5 A steel rule can be read to the nearest millimetre. It is used to measure the length of a bar whose true length is 895 mm. Repeated measurements give the following readings.

length / mm	892, 891, 892, 891, 891, 892

Are the readings accurate and precise to within 1 mm?

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	Results are accurate to within 1 mm	Results are precise to within 1 mm
A	no	no
B	no	yes
C	yes	no
D	yes	yes

(1 mark)

6 The Young modulus of the material of a wire is to be found. The Young modulus E is given by the equation below.

$$E = \frac{4Fl}{\pi d^2 x}$$

The wire is extended by a known force and the following measurements are made.

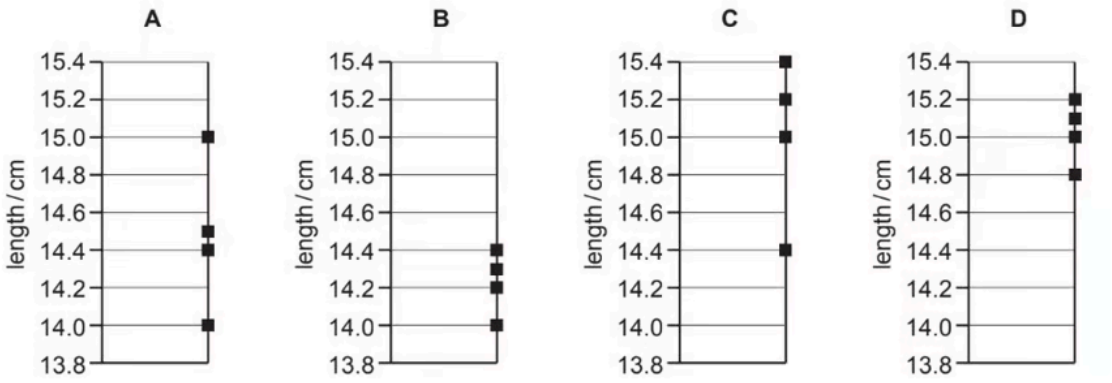
Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	Measurement	Symbol	Value
A	length of wire before force applied	l	2.043 ± 0.002 m
B	diameter of wire	d	0.54 ± 0.02 mm
C	force applied	F	19.62 ± 0.01 N
D	extension of wire with force applied	x	5.2 ± 0.2 mm

(1 mark)

7 Four different students use a ruler to measure the length of a 15.0 cm pencil. Their measurements are recorded on four different charts.

Which chart shows measurements that are precise but **not** accurate?



(1 mark)

8 What is the ratio $\frac{10^2 \text{ THz}}{10^{-4} \text{ pHz}}$?

A. 10^{-30}

B. 10^{-2}

C. 10^{2}

D. 10^{30}

(1 mark)

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Hard Questions

1 A student uses a digital ammeter to measure a current. The reading of the ammeter is found to fluctuate between 1.98 A and 2.02 A.

The manufacturer of the ammeter states that any reading has a systematic uncertainty of ±1%.

Which value of current should be quoted by the student?

A. (2.00 ± 0.01) A

B. (2.00 ± 0.02) A

C. (2.00 ± 0.03) A

D. (2.00 ± 0.04) A

(1 mark)

2 The uncertainty in the value of the momentum of a trolley passing between two points X and Y varies with the choice of measuring devices.

Measurements for the same trolley made by different instruments were recorded.

1	distance between X and Y using a metre rule with cm divisions = 0.55 m
2	distance between X and Y using a metre rule with mm divisions = 0.547 m
3	timings using a wristwatch measuring to the nearest 0.5 s at X = 0.0 s and at Y = 4.5 s
4	timings using light gates measuring to the nearest 0.1 s at X = 0.0s and at Y = 4.3 s
5	mass of trolley using a balance measuring to the nearest g = 6.4×10^{-2} kg
6	mass of trolley using a balance measuring to the nearest 10g = 6×10^{-2} kg

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Which measurements, one for each quantity measured, leads to the least uncertainty in the value of the momentum of the trolley?

A. 1, 3 and 6

B. 1, 4 and 6

C. 2, 3 and 6

D. 2, 4 and 5

(1 mark)

3 In an experiment to determine the acceleration of free fall *g*, a ball bearing is held by an electromagnet. When the current to the electromagnet is switched off, a clock starts and the ball bearing falls. After falling a distance *h*, the ball bearing strikes a switch to stop the clock which measures the time *t* of the fall.

If systematic errors cause *t* and *h* to be measured incorrectly, which error **must** cause *g* to appear greater than 9.81 m s^{-2}?

A. *h* measured as being **smaller** than it actually is and *t* is measured correctly

B. *h* measured as being **smaller** than it actually is and *t* measured as being **larger** than it actually is

C. *h* measured as being **larger** than it actually is and *t* measured as being **larger** than it actually is

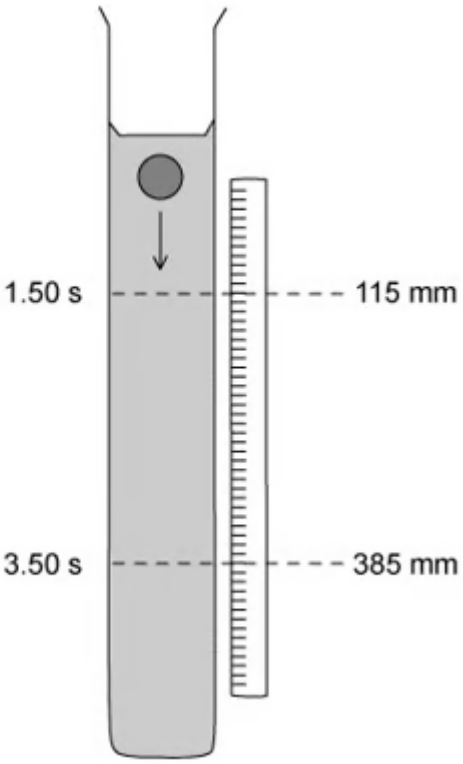
D. *h* is measured correctly and *t* measured as being **smaller** than it actually is

(1 mark)

4 The diagram shows an experiment to measure the speed of a small ball falling at constant

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speed through a clear liquid in a glass tube.



There are two marks on the tube. The top mark is positioned at 115 ± 1 mm on the adjacent rule and the lower mark at 385 ± 1 mm. The ball passes the top mark at 1.50 ± 0.02 s and passes the lower mark at 3.50 ± 0.02 s.

The constant speed of the ball is calculated to be 135 mm s^{-1}

Which expression calculates the fractional uncertainty in the value of this speed?

A. $\frac{2}{270} + \frac{0.04}{2.00}$

B. $\frac{1}{270} + \frac{0.02}{2.00}$

C. $\frac{1}{115} + \frac{0.02}{1.50}$

D. $\frac{1}{385} + \frac{0.02}{3.50}$

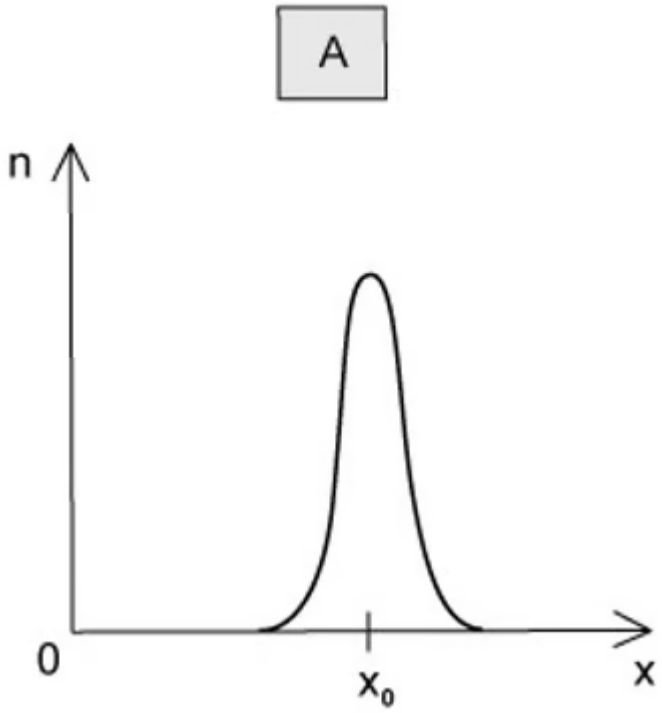
(1 mark)

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5 A fixed quantity x_0 is measured many times in an experiment that has experimental uncertainty. A graph is plotted to show the number n of times that a particular value x is obtained.

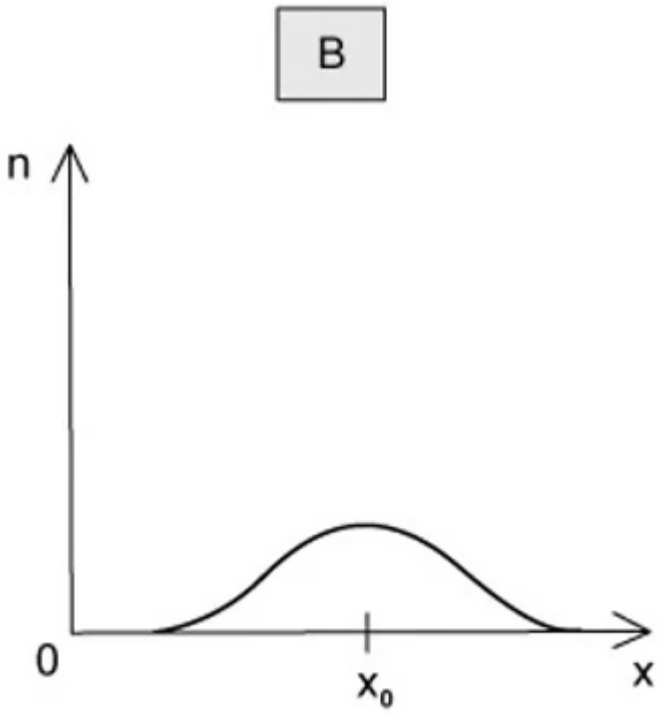
Which graph could be obtained if the measurement of x_0 has a large systematic error but a small random error?

A.

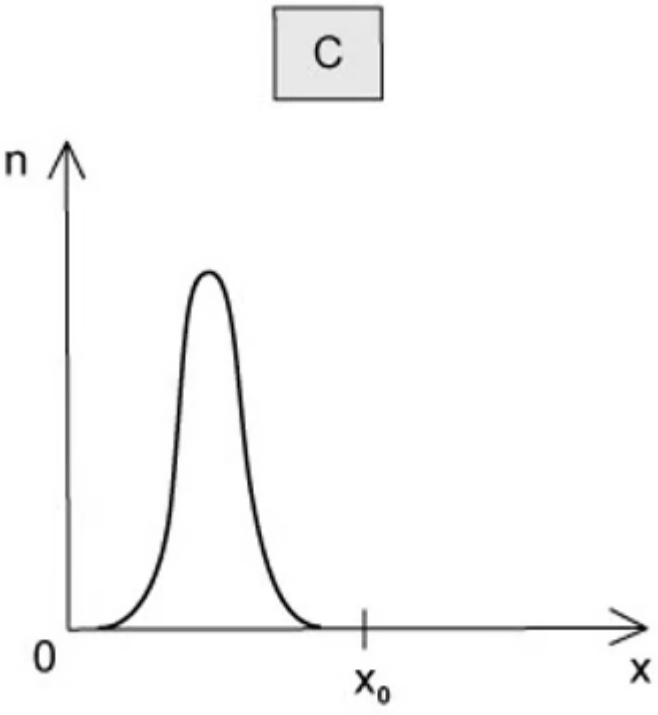


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B.

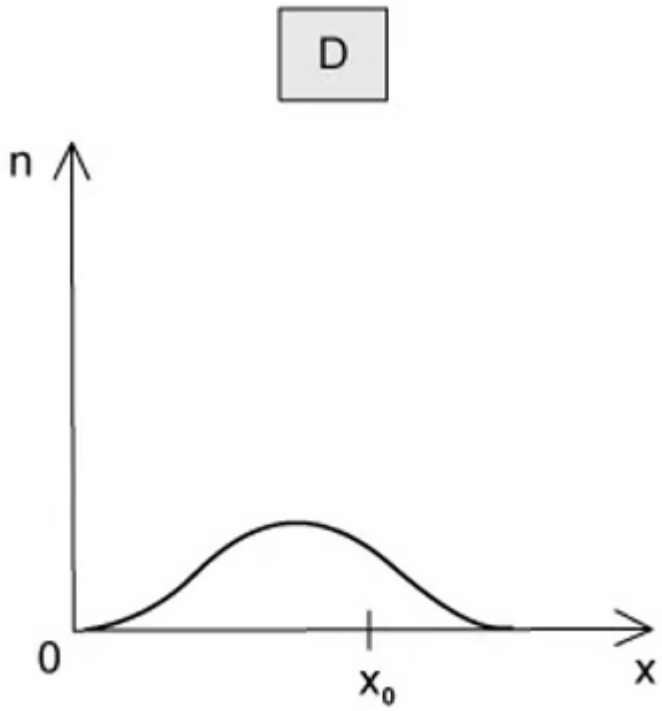


C.



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D.



(1 mark)

6 A student finds the density of a liquid by measuring its mass and volume. The following is a summary of his measurements.

mass of empty beaker = (20 ± 1) g

mass of beaker + liquid = (70 ± 1) g

volume of liquid = (10.0 ± 0.5) cm^3

He correctly calculates the density of the liquid as 5.0 g cm^{-3}.

What is the uncertainty in this value?

A. 0.3 g cm^{-3}

B. 0.5 g cm^{-3}

C. 0.6 g cm^{-3}

D. 2.6 g cm^{-3}

(1 mark)

7 A steel wire is stretched in an experiment to determine the Young modulus for steel.

The uncertainties in the measurements are given below.

Measurement	Uncertainty
load on wire	±2%
length of wire	±0.2%
diameter of wire	±1.5%
extension	±1%

What is the percentage uncertainty in the Young modulus?

A. 1.3%

B. 1.8%

C. 4.7%

D. 6.2%

(1 mark)

8 A mass is dropped from rest and falls through a distance of 2.0 m in a vacuum. An observer records the time taken for the mass to fall through this distance using a manually operated stopwatch and repeats the measurements a further two times. The average result of these measured times, displayed in the table below, was used to determine a value for the acceleration of free fall. This was calculated to be 9.8 m s^{-2}.

	First measurement	Second measurement	Third measurement	Average
Time/ s	0.6	0.73	0.59	0.64

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Which statement best relates to the experiment?

A. The measurements are precise and accurate with no evidence of random errors.

B. The measurements are not accurate and not always recorded to the degree of precision of the measuring device but the calculated experimental result is accurate.

C. The measurements are not always recorded to the degree of precision of the measuring device but are accurate. Systematic errors may be present.

D. The range of results shows that there were random errors made but the calculated value is correct so the experiment was successful.

(1 mark)

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